



Solid Tantalum Surface Mount Chip Capacitors TANTAMOUNT™, Molded Case, Low ESR



LINKS TO ADDITIONAL RESOURCES



3D Models



Calculators



Related Documents



Did You Know?



Technical Notes

Effective September 2005, new capacitor ratings will not be added to the 593D series. All new ratings are available in the TR3 series. The TR3 series offers state-of-the-art low ESR for switch mode power supplies and DC/DC converters.

PERFORMANCE / ELECTRICAL CHARACTERISTICS

www.vishay.com/doc?40192

Operating Temperature: -55 °C to +125 °C
(above +85 °C voltage derating is required)

Capacitance Range: 0.47 µF to 680 µF

Capacitance Tolerance: ± 5 %, ± 10 %, ± 20 %

100 % Surge Current Tested (C, D and E Case Sizes)

Voltage Rating: 4 V_{DC} to 50 V_{DC}

Note

- For recommended voltage derating guidelines see "Typical Performance Characteristics"

FEATURES

- Low ESR
- Molded case available in five case codes
- Terminations: 100 % matte tin standard, tin / lead available
- High ripple current carrying capability
- Compatible with "high volume" automatic pick and place equipment
- Qualified to EIA-717
- Moisture sensitivity level 1
- Compliant terminations
- Standard EIA-535-BAAC case sizes (A through E)
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Note

* This datasheet provides information about parts that are RoHS-compliant and / or parts that are non RoHS-compliant. For example, parts with lead (Pb) terminations are not RoHS-compliant. Please see the information / tables in this datasheet for details



RoHS*
Available

**HALOGEN
FREE
GREEN**
(5-2008)
Available

APPLICATIONS

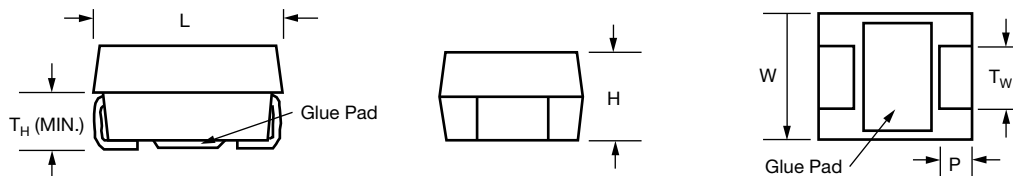
- Industrial
- Telecom infrastructure
- General purpose

ORDERING INFORMATION

593D	107	X9	010	D	2WE3
TYPE	CAPACITANCE	CAPACITANCE TOLERANCE	DC VOLTAGE RATING AT +85 °C	CASE CODE	TERMINATION AND PACKAGING
	This is expressed in picofarads. The first two digits are the significant figures. The third is the number of zeros to follow.	X0 = ± 20 % X9 = ± 10 % X5 = ± 5 % (special order)	This is expressed in V. To complete the three-digit block, zeros precede the voltage rating. A decimal point is indicated by an "R" (6R3 = 6.3 V).	See Ratings and Case Codes table	Matte tin 2TE3 = 7" (178 mm) reel 2WE3 = 13" (330 mm) reel 2DE3 = 7" (178 mm) reel, dry pack 2RE3 = 13" (330 mm) reel, dry pack Tin / lead 8T = 7" (178 mm) reel 8W = 13" (330 mm) reel 8D = 7" (178 mm) reel, dry pack 8R = 13" (330 mm) reel, dry pack

Notes

- We reserve the right to supply higher voltage ratings and tighter capacitance tolerance capacitors in the same case size. Voltage substitutions will be marked with the higher voltage rating.
Low ESR solid tantalum chip capacitors allow delta ESR of 1.25 times the datasheet limits after mounting
- Dry pack as specified in J-STD-033

DIMENSIONS in inches [millimeters]

CASE CODE	EIA SIZE	L	W	H	P	T _W	T _H (MIN.)
A	3216-18	0.126 ± 0.008 [3.2 ± 0.20]	0.063 ± 0.008 [1.6 ± 0.20]	0.063 ± 0.008 [1.6 ± 0.20]	0.031 ± 0.012 [0.80 ± 0.30]	0.047 ± 0.004 [1.2 ± 0.10]	0.028 [0.70]
B	3528-21	0.138 ± 0.008 [3.5 ± 0.20]	0.110 ± 0.008 [2.8 ± 0.20]	0.075 ± 0.008 [1.9 ± 0.20]	0.031 ± 0.012 [0.80 ± 0.30]	0.087 ± 0.004 [2.2 ± 0.10]	0.028 [0.70]
C	6032-28	0.236 ± 0.012 [6.0 ± 0.30]	0.126 ± 0.012 [3.2 ± 0.30]	0.098 ± 0.012 [2.5 ± 0.30]	0.051 ± 0.012 [1.3 ± 0.30]	0.087 ± 0.004 [2.2 ± 0.10]	0.039 [1.0]
D	7343-31	0.287 ± 0.012 [7.3 ± 0.30]	0.169 ± 0.012 [4.3 ± 0.30]	0.110 ± 0.012 [2.8 ± 0.30]	0.051 ± 0.012 [1.3 ± 0.30]	0.094 ± 0.004 [2.4 ± 0.10]	0.039 [1.0]
E	7343-43	0.287 ± 0.012 [7.3 ± 0.30]	0.169 ± 0.012 [4.3 ± 0.30]	0.157 ± 0.012 [4.0 ± 0.30]	0.051 ± 0.012 [1.3 ± 0.30]	0.094 ± 0.004 [2.4 ± 0.10]	0.039 [1.0]

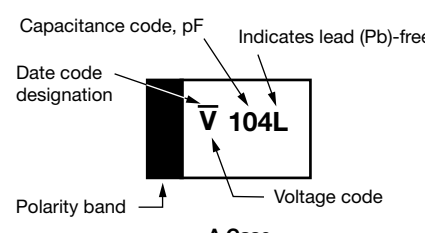
Note

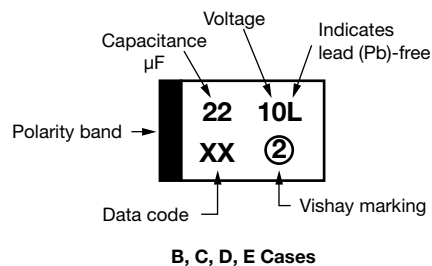
- Glue pad (non-conductive, part of molded case) is dedicated for glue attachment (as user option)

RATINGS AND CASE CODES

μF	4 V	6.3 V	10 V	16 V	20 V	25 V	35 V	50 V
0.47							A	
0.68							A	
1.0				A	A	A	A / B	B / C
1.5						A	B / C	B / C
2.2					A	A / B	B / C	C / D
3.3				A	A	B	C	C / D
4.7			A	A / B	A / B	B / C	C	D / E
6.8			A	A	B	C	C / D	D / E
10		A	A	A / B / C	B / C	C	C / D	D / E
15	A	A	A / B	B / C	B / C	C / D	D / E	
22	A	A / B	A / B / C	B / C	C / D	D	D / E	
33	A / B	A / B	B / C	B / C / D	C / D	D / E		
47	A / B	B / C	B / C / D	C / D	D / E	E		
68	B / C	B / C	C / D	D	D / E			
100	B / C	B / C / D	C / D	D / E	E			
150	B / C / D	C / D / E	D / E	E				
220	C / D	D / E	D / E					
330	D	D / E	E					
470	D / E	E						
680	E							

MARKING

“A” CASE VOLTAGE CODE		 A Case
VOLTS	CODE	
4.0	G	
6.3	J	
10	A	
16	C	
20	D	
25	E	
35	V	
50	T	

**Marking**

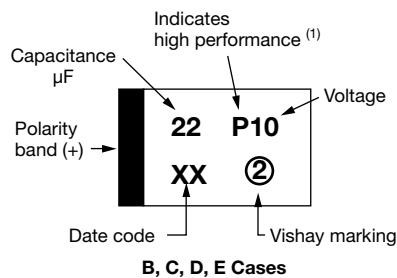
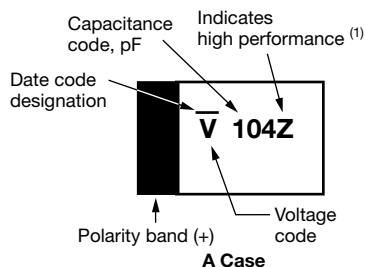
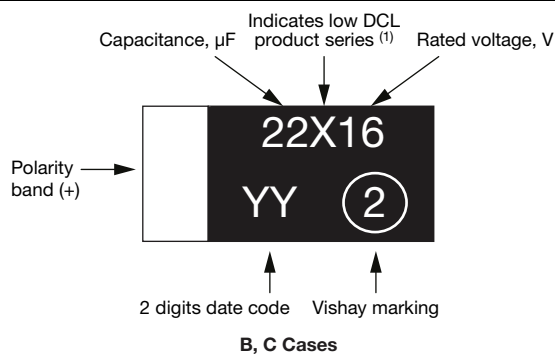
Capacitor marking includes an anode (+) polarity band, capacitance in microfarads and the voltage rating. “A” case capacitors use a letter code for the voltage and EIA capacitance code.

The Vishay identification marking is included if space permits. Vishay marking (“circled 2”) may show additives in the form of short lines, depicting actual manufacturing facility. For A case capacitors discontinuation in polarity bar maybe used as actual manufacturing facility designation. Capacitors rated at 6.3 V are marked 6 V.

A manufacturing date code is marked on all capacitors, for details see FAQ: www.vishay.com/doc?40110.

Capacitors may bear different marking scheme if parts are substituted with high performance Vishay Automotive Grade TP3 family or low DCL TX3 family products. Marking examples are shown below.

Call the factory for further explanation.

ADDITIONAL MARKING EXAMPLES**TP3 MARKING****TX3 MARKING****Note**

(1) Capital letter indicates lead (Pb)-free



STANDARD RATINGS						
CAPACITANCE (μ F)	CASE CODE	PART NUMBER	MAX. DCL AT +25 °C (μ A)	MAX. DF AT +25 °C 120 Hz (%)	MAX. ESR AT +25 °C 100 kHz (Ω)	MAX. RIPPLE 100 kHz I_{RMS} (A)
4 V _{DC} AT +85 °C; 2.7 V _{DC} AT +125 °C						
15	A	593D156(1)004A(2)	0.6	6	1.500	0.22
22	A	593D226(1)004A(2)	0.9	6	1.500	0.22
33	A	593D336(1)004A(2)	1.3	6	1.500	0.22
33	B	593D336(1)004B(2)	1.3	6	0.500	0.41
47	A	593D476(1)004A(2)	1.9	14	0.800	0.31
47	B	593D476(1)004B(2)	1.9	6	0.500	0.41
68	B	593D686(1)004B(2)	2.7	6	0.500	0.41
68	C	593D686(1)004C(2)	2.7	6	0.275	0.63
100	B	593D107(1)004B(2)	4.0	8	0.450	0.43
100	C	593D107(1)004C(2)	4.0	6	0.225	0.66
150	B	593D157(1)004B(2)	6.0	14	0.500	0.41
150	C	593D157(1)004C(2)	6.0	12	0.250	0.66
150	D	593D157(1)004D(2)	6.0	8	0.150	1.00
220	C	593D227(1)004C(2)	8.8	8	0.200	0.74
220	D	593D227(1)004D(2)	8.8	8	0.150	1.00
330	D	593D337(1)004D(2)	13.2	8	0.150	1.00
470	D	593D477(1)004D(2)	18.8	10	0.125	1.10
470	E	593D477(1)004E(2)	18.8	10	0.100	1.28
680	E	593D687(1)004E(2)	27.2	12	0.100	1.28
6.3 V _{DC} AT +85 °C; 4 V _{DC} AT +125 °C						
10	A	593D106(1)6R3A(2)	0.6	6	2.000	0.19
15	A	593D156(1)6R3A(2)	0.9	6	2.000	0.19
22	A	593D226(1)6R3A(2)	1.3	6	2.000	0.19
22	B	593D226(1)6R3B(2)	1.3	6	0.600	0.38
33	A	593D336(1)6R3A(2)	2.0	14	0.800	0.31
33	B	593D336(1)6R3B(2)	2.0	6	0.600	0.38
47	B	593D476(1)6R3B(2)	2.8	6	0.550	0.39
47	C	593D476(1)6R3C(2)	2.8	6	0.300	0.61
68	B	593D686(1)6R3B(2)	4.1	6	0.550	0.39
68	C	593D686(1)6R3C(2)	4.1	6	0.275	0.63
100	B	593D107(1)6R3B(2)	6.0	15	0.500	0.41
100	C	593D107(1)6R3C(2)	6.0	6	0.250	0.66
100	D	593D107(1)6R3D(2)	6.0	6	0.140	1.04
150	C	593D157(1)6R3C(2)	9.0	8	0.200	0.74
150	D	593D157(1)6R3D(2)	9.0	8	0.125	1.10
150	E	593D157(1)6R3E(2)	9.0	8	0.100	1.28
220	D	593D227(1)6R3D(2)	13.2	8	0.100	1.22
220	E	593D227(1)6R3E(2)	13.2	8	0.100	1.28
330	D	593D337(1)6R3D(2)	19.8	8	0.125	1.10
330	E	593D337(1)6R3E(2)	19.8	8	0.100	1.28
470	E	593D477(1)6R3E(2)	28.2	10	0.100	1.28

Note

- Part number definitions:
 - (1) Tolerance: X0, X9. For $\pm 5\%$ tolerance (code X5) contact factory
 - (2) Terminations and packaging: 2TE3, 2WE3, 8T, 8W, 2DE3, 2RE3, 8D, 8R



STANDARD RATINGS						
CAPACITANCE (μ F)	CASE CODE	PART NUMBER	MAX. DCL AT +25 °C (μ A)	MAX. DF AT +25 °C 120 Hz (%)	MAX. ESR AT +25 °C 100 kHz (Ω)	MAX. RIPPLE 100 kHz I_{RMS} (A)
10 V _{DC} AT +85 °C; 7 V _{DC} AT +125 °C						
4.7	A	593D475(1)010A(2)	0.5	6	3.000	0.16
6.8	A	593D685(1)010A(2)	0.7	6	3.000	0.16
10	A	593D106(1)010A(2)	1.0	6	2.000	0.19
15	A	593D156(1)010A(2)	1.5	6	2.000	0.19
15	B	593D156(1)010B(2)	1.5	6	0.700	0.35
22	A	593D226(1)010A(2)	2.2	8	1.500	0.22
22	B	593D226(1)010B(2)	2.2	6	0.700	0.35
22	C	593D226(1)010C(2)	2.2	6	0.345	0.56
33	B	593D336(1)010B(2)	3.3	6	0.600	0.38
33	C	593D336(1)010C(2)	3.3	6	0.300	0.61
47	B	593D476(1)010B(2)	4.7	6	0.600	0.38
47	C	593D476(1)010C(2)	4.7	6	0.300	0.61
47	D	593D476(1)010D(2)	4.7	6	0.200	0.87
68	C	593D686(1)010C(2)	6.8	6	0.275	0.63
68	D	593D686(1)010D(2)	6.8	6	0.150	1.00
100	C	593D107(1)010C(2)	10.0	8	0.200	0.74
100	D	593D107(1)010D(2)	10.0	6	0.100	1.22
150	D	593D157(1)010D(2)	15.0	8	0.100	1.22
150	E	593D157(1)010E(2)	15.0	8	0.100	1.28
220	D	593D227(1)010D(2)	22.0	8	0.125	1.10
220	E	593D227(1)010E(2)	22.0	8	0.100	1.28
330	E	593D337(1)010E(2)	33.0	10	0.100	1.28
16 V _{DC} AT +85 °C; 10 V _{DC} AT +125 °C						
1.0	A	593D105(1)016A(2)	0.5	4	5.500	0.12
3.3	A	593D335(1)016A(2)	0.5	6	3.500	0.15
4.7	A	593D475(1)016A(2)	0.8	6	2.500	0.17
4.7	B	593D475(1)016B(2)	0.8	6	1.500	0.24
6.8	A	593D685(1)016A(2)	1.1	6	3.000	0.16
10	A	593D106(1)016A(2)	1.6	6	1.700	0.21
10	B	593D106(1)016B(2)	1.6	6	0.800	0.33
10	C	593D106(1)016C(2)	1.6	6	0.450	0.49
15	B	593D156(1)016B(2)	2.4	6	0.800	0.33
15	C	593D156(1)016C(2)	2.4	6	0.400	0.52
22	B	593D226(1)016B(2)	3.5	6	0.700	0.35
22	C	593D226(1)016C(2)	3.5	6	0.350	0.56
33	B	593D336(1)016B(2)	5.3	6	0.700	0.35
33	C	593D336(1)016C(2)	5.3	6	0.300	0.61
33	D	593D336(1)016D(2)	4.2	4	0.225	0.82
47	C	593D476(1)016C(2)	7.5	6	0.300	0.61
47	D	593D476(1)016D(2)	7.5	6	0.150	1.00
68	D	593D686(1)016D(2)	10.9	6	0.150	1.00
100	D	593D107(1)016D(2)	16.0	8	0.125	1.10
100	E	593D107(1)016E(2)	16.0	8	0.100	1.28
150	E	593D157(1)016E(2)	24.0	8	0.100	1.28

Note

- Part number definitions:
 - Tolerance: X0, X9. For $\pm 5\%$ tolerance (code X5) contact factory
 - Terminations and packaging: 2TE3, 2WE3, 8T, 8W, 2DE3, 2RE3, 8D, 8R



STANDARD RATINGS						
CAPACITANCE (μ F)	CASE CODE	PART NUMBER	MAX. DCL AT +25 °C (μ A)	MAX. DF AT +25 °C 120 Hz (%)	MAX. ESR AT +25 °C 100 kHz (Ω)	MAX. RIPPLE 100 kHz I_{RMS} (A)
20 V _{DC} AT +85 °C; 13 V _{DC} AT +125 °C						
1.0	A	593D105(1)020A(2)	0.5	4	5.500	0.12
2.2	A	593D225(1)020A(2)	0.5	6	4.000	0.14
3.3	A	593D335(1)020A(2)	0.7	6	4.000	0.14
4.7	A	593D475(1)020A(2)	0.9	6	3.500	0.15
4.7	B	593D475(1)020B(2)	0.9	6	1.000	0.29
6.8	B	593D685(1)020B(2)	1.4	6	1.000	0.29
10	B	593D106(1)020B(2)	2.0	6	1.000	0.29
10	C	593D106(1)020C(2)	2.0	6	0.450	0.49
15	B	593D156(1)020B(2)	3.0	6	1.000	0.29
15	C	593D156(1)020C(2)	3.0	6	0.400	0.52
22	C	593D226(1)020C(2)	4.4	6	0.375	0.54
22	D	593D226(1)020D(2)	3.5	4	0.225	0.82
33	C	593D336(1)020C(2)	6.6	6	0.350	0.56
33	D	593D336(1)020D(2)	6.6	6	0.200	0.87
47	D	593D476(1)020D(2)	9.4	6	0.200	0.87
47	E	593D476(1)020E(2)	7.5	4	0.150	1.05
68	D	593D686(1)020D(2)	13.6	6	0.175	0.93
68	E	593D686(1)020E(2)	13.6	6	0.150	1.05
100	E	593D107(1)020E(2)	20.0	8	0.150	1.05
25 V _{DC} AT +85 °C; 17 V _{DC} AT +125 °C						
1.0	A	593D105(1)025A(2)	0.5	4	4.000	0.14
1.5	A	593D155(1)025A(2)	0.5	6	4.000	0.14
2.2	A	593D225(1)025A(2)	0.6	6	4.000	0.14
2.2	B	593D225(1)025B(2)	0.6	6	1.500	0.24
3.3	B	593D335(1)025B(2)	0.8	6	1.500	0.24
4.7	B	593D475(1)025B(2)	1.2	6	1.500	0.24
4.7	C	593D475(1)025C(2)	1.2	6	0.525	0.46
6.8	C	593D685(1)025C(2)	1.7	6	0.500	0.47
10	C	593D106(1)025C(2)	2.5	6	0.450	0.49
15	C	593D156(1)025C(2)	3.8	6	0.425	0.51
15	D	593D156(1)025D(2)	3.8	6	0.250	0.77
22	D	593D226(1)025D(2)	5.5	6	0.200	0.87
33	D	593D336(1)025D(2)	8.3	6	0.200	0.87
33	E	593D336(1)025E(2)	8.3	6	0.200	0.91
47	E	593D476(1)025E(2)	11.8	6	0.200	0.91

Note

- Part number definitions:
 - (1) Tolerance: X0, X9. For ± 5 % tolerance (code X5) contact factory
 - (2) Terminations and packaging: 2TE3, 2WE3, 8T, 8W, 2DE3, 2RE3, 8D, 8R



STANDARD RATINGS						
CAPACITANCE (μ F)	CASE CODE	PART NUMBER	MAX. DCL AT +25 °C (μ A)	MAX. DF AT +25 °C 120 Hz (%)	MAX. ESR AT +25 °C 100 kHz (Ω)	MAX. RIPPLE 100 kHz I_{RMS} (A)
35 V _{DC} AT +85 °C; 23 V _{DC} AT +125 °C						
0.47	A	593D474(1)035A(2)	0.5	4	4.000	0.14
0.68	A	593D684(1)035A(2)	0.5	4	4.000	0.14
1.0	A	593D105(1)035A(2)	0.5	4	4.000	0.14
1.0	B	593D105(1)035B(2)	0.5	4	2.000	0.21
1.5	B	593D155(1)035B(2)	0.5	6	2.000	0.21
1.5	C	593D155(1)035C(2)	0.5	6	0.900	0.35
2.2	B	593D225(1)035B(2)	0.8	6	2.000	0.21
2.2	C	593D225(1)035C(2)	0.8	6	0.900	0.40
3.3	C	593D335(1)035C(2)	1.2	6	0.700	0.45
4.7	C	593D475(1)035C(2)	1.6	6	0.500	0.47
6.8	C	593D685(1)035C(2)	2.4	6	0.475	0.48
6.8	D	593D685(1)035D(2)	2.4	6	0.300	0.71
10	C	593D106(1)035C(2)	3.5	6	0.450	0.49
10	D	593D106(1)035D(2)	3.5	6	0.300	0.71
15	D	593D156(1)035D(2)	5.3	6	0.300	0.71
15	E	593D156(1)035E(2)	5.3	6	0.300	0.74
22	D	593D226(1)035D(2)	7.7	6	0.300	0.71
22	E	593D226(1)035E(2)	7.7	6	0.275	0.77
50 V _{DC} AT +85 °C; 33 V _{DC} AT +125 °C						
1.0	B	593D105(1)050B(2)	0.5	4	2.000	0.21
1.0	C	593D105(1)050C(2)	0.5	4	1.600	0.26
1.5	B	593D155(1)050B(2)	0.8	6	2.000	0.21
1.5	C	593D155(1)050C(2)	0.8	6	1.500	0.27
2.2	C	593D225(1)050C(2)	1.1	6	1.500	0.27
2.2	D	593D225(1)050D(2)	1.1	6	0.800	0.43
3.3	C	593D335(1)050C(2)	1.7	6	1.500	0.27
3.3	D	593D335(1)050D(2)	1.7	6	0.800	0.43
4.7	D	593D475(1)050D(2)	2.4	6	0.600	0.50
4.7	E	593D475(1)050E(2)	1.9	6	0.600	0.50
6.8	D	593D685(1)050D(2)	3.4	6	0.600	0.50
6.8	E	593D685(1)050E(2)	3.4	6	0.550	0.55
10	D	593D106(1)050D(2)	5.0	6	0.550	0.52
10	E	593D106(1)050E(2)	5.0	6	0.550	0.55

Note

- Part number definitions:
 - Tolerance: X0, X9. For $\pm 5\%$ tolerance (code X5) contact factory
 - Terminations and packaging: 2TE3, 2WE3, 8T, 8W, 2DE3, 2RE3, 8D, 8R

**POWER DISSIPATION**

CASE CODE	MAXIMUM PERMISSIBLE POWER DISSIPATION AT +25 °C (W) IN FREE AIR
A	0.075
B	0.085
C	0.110
D	0.150
E	0.165

STANDARD PACKAGING QUANTITY

CASE CODE	UNITS PER REEL	
	7" REEL	13" REEL
A	2000	9000
B	2000	8000
C	500	3000
D	500	2500
E	400	1500

PRODUCT INFORMATION

Guide for Molded Tantalum Capacitors	www.vishay.com/doc?40074
Pad Dimensions	
Packaging Dimensions	
Moisture Sensitivity (MSL)	www.vishay.com/doc?40135
SELECTOR GUIDES	
Solid Tantalum Selector Guide	www.vishay.com/doc?49053
Solid Tantalum Chip Capacitors	www.vishay.com/doc?40091
FAQ	
Frequently Asked Questions	www.vishay.com/doc?40110



Guide for Molded Tantalum Capacitors

INTRODUCTION

Tantalum electrolytic capacitors are the preferred choice in applications where volumetric efficiency, stable electrical parameters, high reliability, and long service life are primary considerations. The stability and resistance to elevated temperatures of the tantalum / tantalum oxide / manganese dioxide system make solid tantalum capacitors an appropriate choice for today's surface-mount assembly technology.

Vishay Sprague has been a pioneer and leader in this field, producing a large variety of tantalum capacitor types for consumer, industrial, automotive, military, and aerospace electronic applications.

Tantalum is not found in its pure state. Rather, it is commonly found in a number of oxide minerals, often in combination with Columbium ore. This combination is known as "tantalite" when its contents are more than one-half tantalum. Important sources of tantalite include Australia, Brazil, Canada, China, and several African countries. Synthetic tantalite concentrates produced from tin slags in Thailand, Malaysia, and Brazil are also a significant raw material for tantalum production.

Electronic applications, and particularly capacitors, consume the largest share of world tantalum production. Other important applications for tantalum include cutting tools (tantalum carbide), high temperature super alloys, chemical processing equipment, medical implants, and military ordnance.

Vishay Sprague is a major user of tantalum materials in the form of powder and wire for capacitor elements and rod and sheet for high temperature vacuum processing.

THE BASICS OF TANTALUM CAPACITORS

Most metals form crystalline oxides which are non-protecting, such as rust on iron or black oxide on copper. A few metals form dense, stable, tightly adhering, electrically insulating oxides. These are the so-called "valve" metals and include titanium, zirconium, niobium, tantalum, hafnium, and aluminum. Only a few of these permit the accurate control of oxide thickness by electrochemical means. Of these, the most valuable for the electronics industry are aluminum and tantalum.

Capacitors are basic to all kinds of electrical equipment, from radios and television sets to missile controls and automobile ignitions. Their function is to store an electrical charge for later use.

Capacitors consist of two conducting surfaces, usually metal plates, whose function is to conduct electricity. They are separated by an insulating material or dielectric. The dielectric used in all tantalum electrolytic capacitors is tantalum pentoxide.

Tantalum pentoxide compound possesses high-dielectric strength and a high-dielectric constant. As capacitors are being manufactured, a film of tantalum pentoxide is applied to their electrodes by means of an electrolytic process. The film is applied in various thicknesses and at various voltages and although transparent to begin with, it takes on different colors as light refracts through it. This coloring occurs on the tantalum electrodes of all types of tantalum capacitors.

Rating for rating, tantalum capacitors tend to have as much as three times better capacitance / volume efficiency than aluminum electrolytic capacitors. An approximation of the capacitance / volume efficiency of other types of capacitors may be inferred from the following table, which shows the dielectric constant ranges of the various materials used in each type. Note that tantalum pentoxide has a dielectric constant of 26, some three times greater than that of aluminum oxide. This, in addition to the fact that extremely thin films can be deposited during the electrolytic process mentioned earlier, makes the tantalum capacitor extremely efficient with respect to the number of microfarads available per unit volume. The capacitance of any capacitor is determined by the surface area of the two conducting plates, the distance between the plates, and the dielectric constant of the insulating material between the plates.

COMPARISON OF CAPACITOR DIELECTRIC CONSTANTS

DIELECTRIC	ϵ DIELECTRIC CONSTANT
Air or vacuum	1.0
Paper	2.0 to 6.0
Plastic	2.1 to 6.0
Mineral oil	2.2 to 2.3
Silicone oil	2.7 to 2.8
Quartz	3.8 to 4.4
Glass	4.8 to 8.0
Porcelain	5.1 to 5.9
Mica	5.4 to 8.7
Aluminum oxide	8.4
Tantalum pentoxide	26
Ceramic	12 to 400K

In the tantalum electrolytic capacitor, the distance between the plates is very small since it is only the thickness of the tantalum pentoxide film. As the dielectric constant of the tantalum pentoxide is high, the capacitance of a tantalum capacitor is high if the area of the plates is large:

$$C = \frac{\epsilon A}{t}$$

where

C = capacitance

ϵ = dielectric constant

A = surface area of the dielectric

t = thickness of the dielectric

Tantalum capacitors contain either liquid or solid electrolytes. In solid electrolyte capacitors, a dry material (manganese dioxide) forms the cathode plate. A tantalum lead is embedded in or welded to the pellet, which is in turn connected to a termination or lead wire. The drawings show the construction details of the surface-mount types of tantalum capacitors shown in this catalog.

SOLID ELECTROLYTE TANTALUM CAPACITORS

Solid electrolyte capacitors contain manganese dioxide, which is formed on the tantalum pentoxide dielectric layer by impregnating the pellet with a solution of manganous nitrate. The pellet is then heated in an oven, and the manganous nitrate is converted to manganese dioxide.

The pellet is next coated with graphite, followed by a layer of metallic silver, which provides a conductive surface between the pellet and the leadframe.

Molded Chip tantalum capacitor encases the element in plastic resins, such as epoxy materials. The molding compound has been selected to meet the requirements of UL 94 V-0 and outgassing requirements of ASTM E-595. After assembly, the capacitors are tested and inspected to assure long life and reliability. It offers excellent reliability and high stability for consumer and commercial electronics with the added feature of low cost

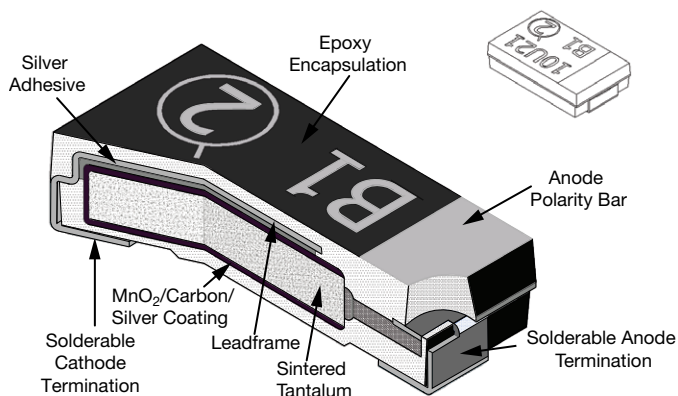
Surface-mount designs of “solid tantalum” capacitors use lead frames or lead frameless designs as shown in the accompanying drawings.

TANTALUM CAPACITORS FOR ALL DESIGN CONSIDERATIONS

Solid electrolyte designs are the least expensive for a given rating and are used in many applications where their very small size for a given unit of capacitance is of importance. They will typically withstand up to about 10 % of the rated DC working voltage in a reverse direction. Also important are their good low temperature performance characteristics and freedom from corrosive electrolytes.

Vishay Sprague patented the original solid electrolyte capacitors and was the first to market them in 1956. Vishay Sprague has the broadest line of tantalum capacitors and has continued its position of leadership in this field. Data sheets covering the various types and styles of Vishay Sprague capacitors for consumer and entertainment electronics, industry, and military applications are available where detailed performance characteristics must be specified.

MOLDED CHIP CAPACITOR



COMMERCIAL PRODUCTS

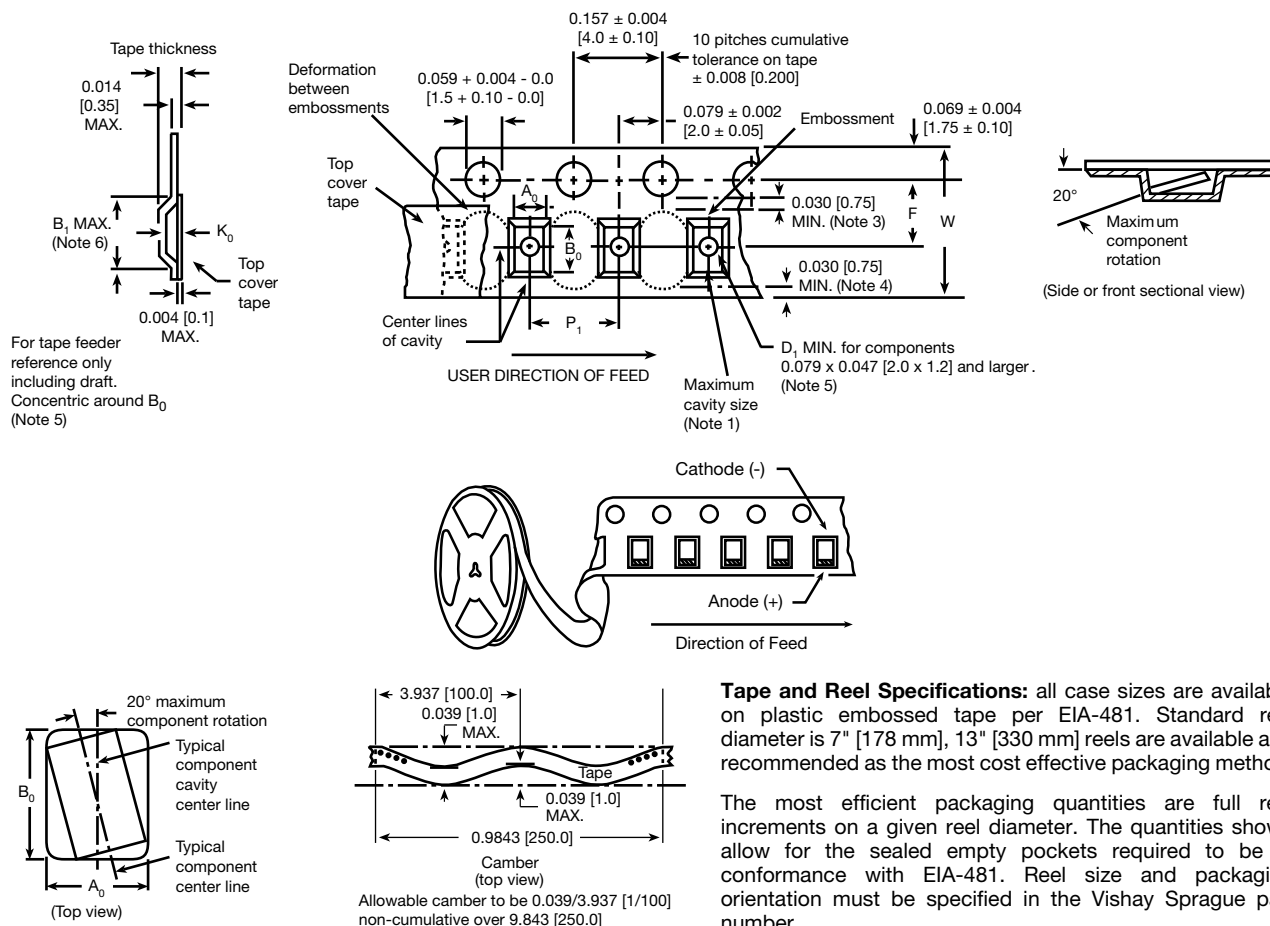
SOLID TANTALUM CAPACITORS - MOLDED CASE						
SERIES	293D	793DX-CTC3-CTC4	593D	TR3	TP3	TL3
PRODUCT IMAGE						
TYPE	Surface-mount TANTAMOUNT™, molded case					
FEATURES	Standard industrial grade	CECC approved	Low ESR	Low ESR	High performance, automotive grade	Very low DCL
TEMPERATURE RANGE	-55 °C to +125 °C					
CAPACITANCE RANGE	0.1 µF to 1000 µF	0.1 µF to 100 µF	1 µF to 470 µF	0.47 µF to 1000 µF	0.1 µF to 470 µF	0.1 µF to 470 µF
VOLTAGE RANGE	4 V to 75 V	4 V to 50 V	4 V to 50 V	4 V to 75 V	4 V to 50 V	4 V to 50 V
CAPACITANCE TOLERANCE	± 10 %, ± 20 %					
LEAKAGE CURRENT	0.01 CV or 0.5 µA, whichever is greater					0.005 CV or 0.25 µA, whichever is greater
DISSIPATION FACTOR	4 % to 30 %	4 % to 6 %	4 % to 15 %	4 % to 30 %	4 % to 15 %	4 % to 15 %
CASE CODES	A, B, C, D, E	A, B, C, D	A, B, C, D, E	A, B, C, D, E, W	A, B, C, D, E	A, B, C, D, E
TERMINATION	100 % matte tin standard, tin / lead available					

SOLID TANTALUM CAPACITORS - MOLDED CASE				
SERIES	TX3	TH3	TH4	TH5
PRODUCT IMAGE				
TYPE	Surface-mount TANTAMOUNT™, molded case			
FEATURES	E-detectors	High temperature +150 °C, automotive grade	High temperature +175 °C, automotive grade	Very high temperature +200 °C
TEMPERATURE RANGE	-55 °C to +125 °C	-55 °C to +150 °C	-55 °C to +175 °C	-55 °C to +200 °C
CAPACITANCE RANGE	10 µF to 100 µF	0.33 µF to 220 µF	10 µF to 100 µF	4.7 µF to 100 µF
VOLTAGE RANGE	16 V to 25 V	6.3 V to 50 V	6.3 V to 35 V	5 V to 24 V
CAPACITANCE TOLERANCE	± 10 %, ± 20 %			
LEAKAGE CURRENT	0.005 CV	0.01 CV or 0.5 µA, whichever is greater		
DISSIPATION FACTOR	6 % to 20 %	4 % to 8 %	4.5 % to 8 %	6 % to 10 %
CASE CODES	B, C	A, B, C, D, E	B, C, D, E	D, E
TERMINATION	100 % matte tin	100 % matte tin standard, tin / lead and gold plated available	100 % matte tin	Gold plated

HIGH RELIABILITY PRODUCTS

SOLID TANTALUM CAPACITORS - MOLDED CASE				
SERIES	TM3	T83	CWR11	95158
PRODUCT IMAGE				
TYPE	TANTAMOUNT™, molded case, hi-rel.	TANTAMOUNT™, molded case, hi-rel. COTS	TANTAMOUNT™, molded case, DLA approved	
FEATURES	High reliability, for medical Instruments	High reliability, standard and low ESR	MIL-PRF-55365/8 qualified	Low ESR
TEMPERATURE RANGE	-55 °C to +125 °C			
CAPACITANCE RANGE	1 µF to 220 µF	0.1 µF to 470 µF	0.1 µF to 100 µF	4.7 µF to 220 µF
VOLTAGE RANGE	4 V to 20 V	4 V to 63 V	4 V to 50 V	
CAPACITANCE TOLERANCE	± 10 %, ± 20 %		± 5 %, ± 10 %, ± 20 %	± 10 %, ± 20 %
LEAKAGE CURRENT	0.005 CV or 0.25 µA, whichever is greater	0.01 CV or 0.5 µA, whichever is greater		
DISSIPATION FACTOR	4 % to 8 %	4 % to 15 %	4 % to 6 %	4 % to 12 %
CASE CODES	A, B, C, D, E	A, B, C, D, E	A, B, C, D	C, D, E
TERMINATION	100 % matte tin; tin / lead	100 % matte tin; tin / lead; tin / lead solder fused	Tin / lead; tin / lead solder fused	Tin / lead solder plated; gold plated

PLASTIC TAPE AND REEL PACKAGING in inches [millimeters]



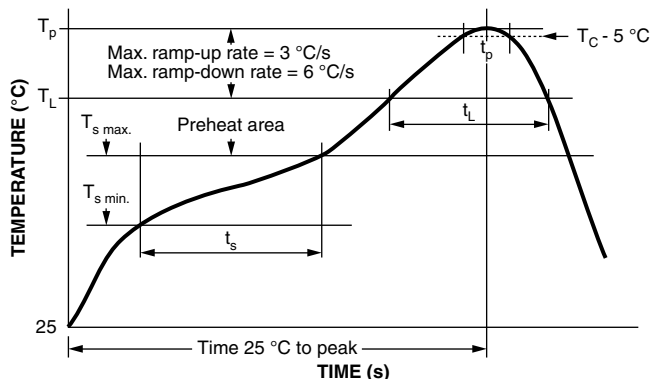
Notes

- Metric dimensions will govern. Dimensions in inches are rounded and for reference only
- A_0 , B_0 , K_0 are determined by the maximum dimensions to the ends of the terminals extending from the component body and / or the body dimensions of the component. The clearance between the ends of the terminals or body of the component to the sides and depth of the cavity (A_0 , B_0 , K_0) must be within 0.002" (0.05 mm) minimum and 0.020" (0.50 mm) maximum. The clearance allowed must also prevent rotation of the component within the cavity of not more than 20°
- Tape with components shall pass around radius "R" without damage. The minimum trailer length may require additional length to provide "R" minimum for 12 mm embossed tape for reels with hub diameters approaching N minimum
- This dimension is the flat area from the edge of the sprocket hole to either outward deformation of the carrier tape between the embossed cavities or to the edge of the cavity whichever is less
- This dimension is the flat area from the edge of the carrier tape opposite the sprocket holes to either the outward deformation of the carrier tape between the embossed cavity or to the edge of the cavity whichever is less
- The embossed hole location shall be measured from the sprocket hole controlling the location of the embossment. Dimensions of embossment location shall be applied independent of each other
- B_1 dimension is a reference dimension tape feeder clearance only

CASE CODE	TAPE SIZE	B_1 (MAX.)	D_1 (MIN.)	F	K_0 (MAX.)	P_1	W
MOLDED CHIP CAPACITORS; ALL TYPES							
A	8 mm	0.165	0.039	0.138 ± 0.002	0.094	0.157 ± 0.004	0.315 ± 0.012
B		[4.2]	[1.0]	[3.5 ± 0.05]	[2.4]	[4.0 ± 1.0]	[8.0 ± 0.30]
C							
D	12 mm	0.32	0.059	0.217 ± 0.00	0.177	0.315 ± 0.004	0.472 ± 0.012
E		[8.2]	[1.5]	[5.5 ± 0.05]	[4.5]	[8.0 ± 1.0]	[12.0 ± 0.30]
W							

RECOMMENDED REFLOW PROFILES

Capacitors should withstand reflow profile as per J-STD-020 standard, three cycles.

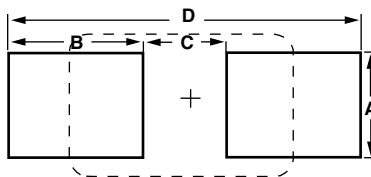


PROFILE FEATURE	SnPb EUTECTIC ASSEMBLY	LEAD (Pb)-FREE ASSEMBLY
Preheat / soak		
Temperature min. ($T_{s \text{ min.}}$)	100 °C	150 °C
Temperature max. ($T_{s \text{ max.}}$)	150 °C	200 °C
Time (t_s) from ($T_{s \text{ min.}}$ to $T_{s \text{ max.}}$)	60 s to 120 s	60 s to 120 s
Ramp-up		
Ramp-up rate (T_L to T_p)	3 °C/s max.	3 °C/s max.
Liquidus temperature (T_L)	183 °C	217 °C
Time (t_L) maintained above T_L	60 s to 150 s	60 s to 150 s
Peak package body temperature (T_p)	Depends on case size - see table below	
Time (t_p) within 5 °C of the specified classification temperature (T_C)	20 s	30 s
Time 25 °C to peak temperature	6 min max.	8 min max.
Ramp-down		
Ramp-down rate (T_p to T_L)	6 °C/s max.	6 °C/s max.

PEAK PACKAGE BODY TEMPERATURE (T_p)

CASE CODE	PEAK PACKAGE BODY TEMPERATURE (T_p)	
	SnPb EUTECTIC PROCESS	LEAD (Pb)-FREE PROCESS
A, B, C	235 °C	260 °C
D, E, W	220 °C	250 °C

PAD DIMENSIONS in inches [millimeters]



CASE CODE	A (MIN.)	B (NOM.)	C (NOM.)	D (NOM.)
MOLDED CHIP CAPACITORS, ALL TYPES				
A	0.071 [1.80]	0.067 [1.70]	0.053 [1.35]	0.187 [4.75]
B	0.118 [3.00]	0.071 [1.80]	0.065 [1.65]	0.207 [5.25]
C	0.118 [3.00]	0.094 [2.40]	0.118 [3.00]	0.307 [7.80]
D	0.157 [4.00]	0.098 [2.50]	0.150 [3.80]	0.346 [8.80]
E	0.157 [4.00]	0.098 [2.50]	0.150 [3.80]	0.346 [8.80]
W	0.185 [4.70]	0.098 [2.50]	0.150 [3.80]	0.346 [8.80]



GUIDE TO APPLICATION

1. **AC Ripple Current:** the maximum allowable ripple current shall be determined from the formula:

$$I_{RMS} = \sqrt{\frac{P}{R_{ESR}}}$$

where,

P = power dissipation in W at +25 °C as given in the tables in the product datasheets (Power Dissipation).

R_{ESR} = the capacitor equivalent series resistance at the specified frequency

2. **AC Ripple Voltage:** the maximum allowable ripple voltage shall be determined from the formula:

$$V_{RMS} = I_{RMS} \times Z$$

or, from the formula:

$$V_{RMS} = Z \sqrt{\frac{P}{R_{ESR}}}$$

where,

P = power dissipation in W at +25 °C as given in the tables in the product datasheets (Power Dissipation).

R_{ESR} = the capacitor equivalent series resistance at the specified frequency

Z = the capacitor impedance at the specified frequency

- 2.1 The sum of the peak AC voltage plus the applied DC voltage shall not exceed the DC voltage rating of the capacitor.
- 2.2 The sum of the negative peak AC voltage plus the applied DC voltage shall not allow a voltage reversal exceeding 10 % of the DC working voltage at +25 °C.
3. **Reverse Voltage:** solid tantalum capacitors are not intended for use with reverse voltage applied. However, they have been shown to be capable of withstanding momentary reverse voltage peaks of up to 10 % of the DC rating at 25 °C and 5 % of the DC rating at +85 °C.
4. **Temperature Derating:** if these capacitors are to be operated at temperatures above +25 °C, the permissible RMS ripple current shall be calculated using the derating factors as shown:

TEMPERATURE (°C)	DERATING FACTOR
+25	1.0
+85	0.9
+125	0.4
+150 ⁽¹⁾	0.3
+175 ⁽¹⁾	0.2
+200 ⁽¹⁾	0.1

Note

⁽¹⁾ Applicable for dedicated high temperature product series

5. **Power Dissipation:** power dissipation will be affected by the heat sinking capability of the mounting surface. Non-sinusoidal ripple current may produce heating effects which differ from those shown. It is important that the equivalent I_{RMS} value

be established when calculating permissible operating levels. (Power dissipation calculated using +25 °C temperature rise).

6. **Printed Circuit Board Materials:** molded capacitors are compatible with commonly used printed circuit board materials (alumina substrates, FR4, FR5, G10, PTFE-fluorocarbon and porcelainized steel).

7. **Attachment:**

- 7.1 **Solder Paste:** the recommended thickness of the solder paste after application is 0.007" ± 0.001" [0.178 mm ± 0.025 mm]. Care should be exercised in selecting the solder paste. The metal purity should be as high as practical. The flux (in the paste) must be active enough to remove the oxides formed on the metallization prior to the exposure to soldering heat. In practice this can be aided by extending the solder preheat time at temperatures below the liquidous state of the solder.

- 7.2 **Soldering:** capacitors can be attached by conventional soldering techniques; vapor phase, convection reflow, infrared reflow, wave soldering, and hot plate methods. The soldering profile charts show recommended time / temperature conditions for soldering. Preheating is recommended. The recommended maximum ramp rate is 3 °C per second. Attachment with a soldering iron is not recommended due to the difficulty of controlling temperature and time at temperature. The soldering iron must never come in contact with the capacitor. For details see www.vishay.com/doc?40214.

- 7.2.1 **Backward and Forward Compatibility:** capacitors with SnPb or 100 % tin termination finishes can be soldered using SnPb or lead (Pb)-free soldering processes.

8. **Cleaning (Flux Removal) After Soldering:** molded capacitors are compatible with all commonly used solvents such as TES, TMS, Prelete, Chloroethane, Terpene and aqueous cleaning media. However, CFC / ODS products are not used in the production of these devices and are not recommended. Solvents containing methylene chloride or other epoxy solvents should be avoided since these will attack the epoxy encapsulation material.

- 8.1 When using ultrasonic cleaning, the board may resonate if the output power is too high. This vibration can cause cracking or a decrease in the adherence of the termination. DO NOT EXCEED 9W/l at 40 kHz for 2 min.

9. **Recommended Mounting Pad Geometries:** proper mounting pad geometries are essential for successful solder connections. These dimensions are highly process sensitive and should be designed to minimize component rework due to unacceptable solder joints. The dimensional configurations shown are the recommended pad geometries for both wave and reflow soldering techniques. These dimensions are intended to be a starting point for circuit board designers and may be fine tuned if necessary based upon the peculiarities of the soldering process and / or circuit board design.



Molded Chip Tantalum Capacitors

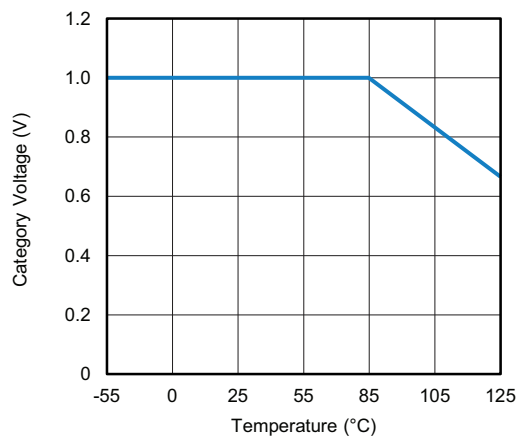
CAPACITOR ELECTRICAL PERFORMANCE CHARACTERISTICS				
ITEM	PERFORMANCE CHARACTERISTICS			
Category temperature range	-55 °C to +85 °C (to +125 °C with voltage derating)			
Capacitance tolerance	± 20 %, ± 10 %. Tested via bridge method, at +25 °C, 120 Hz			
Dissipation factor	Limit per Standard Ratings table. Tested via bridge method, at 25 °C, 120 Hz			
ESR	Limit per Standard Ratings table. Tested via bridge method, at 25 °C, 100 kHz			
Leakage current	After application of rated voltage applied to capacitors for 5 min using a steady source of power with 1 kΩ resistor in series with the capacitor under test, leakage current at 25 °C is not more than 0.01 CV or 0.5 μA, whichever is greater. <i>Note that the leakage current varies with temperature and applied voltage. See graph below for the appropriate adjustment factor.</i>			
Capacitance change by temperature	+20 % max. (at +125 °C) +10 % max. (at +85 °C) -10 % max. (at -55 °C)			
Reverse voltage	Capacitors are capable of withstanding peak voltages in the reverse direction equal to: 10 % of the DC rating at +25 °C 5 % of the DC rating at +85 °C Vishay does not recommend intentional or repetitive application of reverse voltage			
Ripple current	For maximum ripple current values (at 25 °C) refer to relevant datasheet. If capacitors are to be used at temperatures above +25 °C, the permissible RMS ripple current (or voltage) shall be calculated using the derating factors: 1.0 at +25 °C; 0.9 at +85 °C; 0.4 at +125 °C			
Maximum operating and surge voltages vs. temperature	+85 °C		+125 °C	
	RATED VOLTAGE (V)	SURGE VOLTAGE (V)	CATEGORY VOLTAGE (V)	SURGE VOLTAGE (V)
	4	5.2	2.7	3.4
	6.3	8	4	5
	10	13	7	8
	16	20	10	12
	20	26	13	16
	25 ⁽¹⁾	32	17	20
	25 ⁽²⁾	27	17	20
	35 ⁽³⁾	46	23	28
	35 ⁽⁴⁾	42	23	28
	50	65	33	40
	50 ⁽⁵⁾	60	33	40
	63	75	42	50
	75 ⁽⁶⁾	75	50	50
Recommended voltage derating guidelines (below 85 °C) ⁽⁷⁾	VOLTAGE RAIL (V)		CAPACITOR VOLTAGE RATING (V)	
	≤ 3.3		6.3	
	5		10	
	10		20	
	12		25	
	15		35	
	24		50 or series configuration	
	28		63 or series configuration	
	≥ 32		75 or series configuration	

Notes

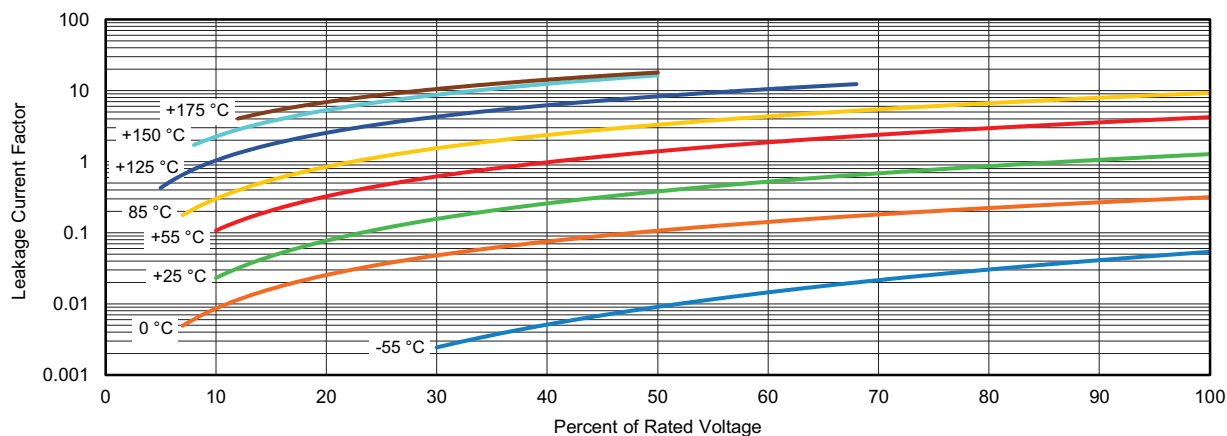
- All information presented in this document reflects typical performance characteristics
 - For more information about recommended voltage derating see: www.vishay.com/doc?40246
- (1) Capacitance values lower than 100 μF
 (2) Capacitance values 100 μF and higher
 (3) Capacitance values lower than 33 μF
 (4) Capacitance values 33 μF and higher
 (5) Capacitance values 15 μF and higher
 (6) For 293D and TR3 only
 (7) For temperatures above +85 °C the same voltage derating ratio is recommended, but with respect to category voltage: up to +85 °C: category voltage = rated voltage; at +125 °C: category voltage = 2/3 of rated voltage, between these temperatures it decreases linearly - see graph below



CATEGORY VOLTAGE VS. TEMPERATURE



TYPICAL LEAKAGE CURRENT TEMPERATURE FACTOR



Note

- At +25 °C, the leakage current shall not exceed the value listed in the Standard Ratings table.
- At +85 °C, the leakage current shall not exceed 10 times the value listed in the Standard Ratings table.
- At +125 °C, the leakage current shall not exceed 12 times the value listed in the Standard Ratings table.
- At +150 °C, the leakage current shall not exceed 15 times the value listed in the Standard Ratings table.
- At +175 °C, the leakage current shall not exceed 18 times the value listed in the Standard Ratings table.

**ENVIRONMENTAL PERFORMANCE CHARACTERISTICS**

ITEM	CONDITION	POST TEST PERFORMANCE	
Surge voltage	Post application of surge voltage (as specified in the table above) in series with a 33 Ω resistor at the rate of 30 s ON, 30 s OFF, for 1000 successive test cycles at 85 °C.	Capacitance change Dissipation factor Leakage current	Within ± 10 % of initial value Initial specified limit Initial specified limit
Life test at +85 °C	1000 h application of rated voltage at 85 °C. MIL-STD-202, method 108	Capacitance change Dissipation factor Leakage current	Within -20 % / +10 % of initial value Initial specified limit Shall not exceed 125 % of initial limit
Life test at +125 °C	1000 h application 2/3 of rated voltage at 125 °C. MIL-STD-202, method 108	Capacitance change Dissipation factor Leakage current	Within -20 % / +10 % of initial value Initial specified limit Shall not exceed 125 % of initial limit
Humidity tests	At 60 °C / 90 % RH 1000 h, biased	Capacitance change Dissipation factor Leakage current	Within -10 % / +20 % of initial value Not to exceed 150 % of initial limit Shall not exceed 200 % of initial limit
Thermal shock	MIL-STD-202, method 107, test condition A (-55 °C / +85 °C, for 1000 cycles)	Capacitance change Dissipation factor Leakage current	Within ± 10 % of initial value Initial specified limit Initial specified limit

MECHANICAL PERFORMANCE CHARACTERISTICS

TEST CONDITION	CONDITION	POST TEST PERFORMANCE	
Terminal strength / shear force test	Apply a pressure load of 17.7 N for 60 s horizontally to the center of capacitor side body.	Capacitance change Dissipation factor Leakage current	Within ± 10 % of initial value Initial specified limit Initial specified limit
		There shall be no mechanical or visual damage to capacitors post-conditioning.	
Vibration	MIL-STD-202, method 204, condition D, 10 Hz to 2000 Hz, 20 g peak, 8 h, at rated voltage	Electrical measurements are not applicable, since the same parts are used for shock (specified pulse) test. There shall be no mechanical or visual damage to capacitors post-conditioning.	
Shock (specified pulse)	MIL-STD-202, method 213, condition I, 100 g peak	Capacitance change Dissipation factor Leakage current	Within ± 10 % of initial value Initial specified limit Initial specified limit
		There shall be no mechanical or visual damage to capacitors post-conditioning.	
Resistance to soldering heat	Recommended reflow profiles temperatures and durations are located within the Capacitor Series Guides MIL-STD-202, method 210, condition B	Capacitance change Dissipation factor Leakage current	Within ± 10 % of initial value Initial specified limit Initial specified limit
		There shall be no mechanical or visual damage to capacitors post-conditioning.	
Solderability and dissolution of metallization	MIL-STD-202, method 208, ANSI/J-STD-002, test B (SnPb) and B1 (lead (Pb)-free). Dissolution of metallization: method D. Does not apply to gold terminations.	There shall be no mechanical or visual damage to capacitors post-conditioning.	
Flammability	Encapsulation materials meet UL 94 V-0 with an oxygen index of 32 %.		



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